4.5 The Sophomore's Dream

WORKSHEET

The question

STEP III 2009, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/09-s3-q8

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The function x^x is not defined at $x = 0$ by any formula you know (0^0 is ambiguous). How does it behave as x approaches 0 from above? Try $x = 0.1, 0.01, 0.001$ on a calculator.

STAGE 1 Powers beat logarithms

Ask yourself As $x \to 0^+$, $\ln x \to -\infty$. What substitution turns $x \to 0^+$ into $t \to \infty$, and turns $x^m(\ln x)^n$ into something like $e^{-mt}t^n$?

STAGE 2 A reduction formula

Ask yourself Integrate $I_{n+1} = \int_0^1 x^m(\ln x)^{n+1}\,dx$ by parts. Which factor should you integrate? What happens to the boundary term at $x = 0$?

STAGE 3 The series

Ask yourself Expand $x^x = e^{x \ln x}$ using the series for e^y. What does each term look like? Which value of m makes the integrals in part (ii) appear?

Pause and try Integrate term by term from 0 to 1, using part (ii) with $m = n$ (and treating $n = 0$ separately). What do you get?

What's really going on

Swapping sums and integrals

Stage 3 quietly swapped an infinite sum with an integral: $\int \sum = \sum \int$. That is not always allowed. At university you will meet examples where it fails, and theorems that say when it is safe. The simplest says that it is safe when the series converges *uniformly*: the tail of the series can be made small at every x simultaneously. Here, $|x \ln x| \le \frac{1}{e}$ on $(0, 1]$, so the nth term is at most $\frac{e^{-n}}{n!}$ in size, whatever x is. The sum of these bounds converges, so the tails are uniformly small and the swap is justified. This argument is called the *Weierstrass M-test*, and it is one of the workhorses of analysis.

Why the "dream" works

The identity is special. It depends on the exact match between the powers in $\frac{(x \ln x)^n}{n!}$ and the integral $\int_0^1 x^n (\ln x)^n \, dx = \frac{(-1)^n n!}{(n+1)^{n+1}}$, where the $n!$ cancels perfectly. Underneath, this integral is a Gamma function in disguise: the substitution $x = e^{-t/(n+1)}$ turns it

into $\frac{(-1)^n}{(n+1)^{n+1}} \int_0^\infty t^n e^{-t}\,dt = \frac{(-1)^n\,\Gamma(n+1)}{(n+1)^{n+1}}$, connecting with Section 4.4.

The name

The name "sophomore's dream" was coined by Jonathan Borwein and his co-authors in their 2004 book *Experimentation in Mathematics*, as a companion to the "freshman's dream" $(a+b)^n = a^n + b^n$ (which, curiously, is true in some number systems: in arithmetic modulo a prime p, $(a+b)^p = a^p + b^p$, because all the middle binomial coefficients are divisible by p). Borwein was a pioneer of *experimental mathematics*, in which computers are used to discover identities like this one, which are then proved by hand.

Try this next

1 **The companion.** Show that $\displaystyle\int_0^1 x^{-x}\,dx = \sum_{n=1}^{\infty} \frac{1}{n^n} \approx 1.2913$.

2 **The minimum.** Show that x^x has its minimum on $(0,\infty)$ at $x = \frac{1}{e}$, with value $e^{-1/e}$.

3 **How many terms?** For an alternating series with decreasing terms, the error after stopping is less than the first omitted term. How many terms of the series for $\int_0^1 x^x\,dx$ are needed for six decimal places?

4 **A cousin.** Use the same method to show that $\displaystyle\int_0^1 x^2 \ln x\,dx = -\frac{1}{9}$, and that $\displaystyle\int_0^1 \frac{\ln x}{1-x}\,dx = -\sum_{n=1}^{\infty} \frac{1}{n^2}$. (By Section 2.4, this is $-\frac{\pi^2}{6}$.)

Further reading

- **Jonathan Borwein, David Bailey & Roland Girgensohn, *Experimentation in Mathematics*** (A. K. Peters). The source of the name, and many more identities discovered by computer.
- **William Dunham, *The Calculus Gallery*** (Princeton University Press). Masterpieces of analysis from Newton to Lebesgue, including Bernoulli's work.
- **Mary Hart, *Guide to Analysis*** (Palgrave). Clear treatment of uniform convergence and when sums and integrals can be swapped.

The full guided solution is in Section 4.5:
https://beyondstep.keithmcnulty.org/ch04/sophomores-dream.html